Information Presentation and Data Visualisation - Lab 01

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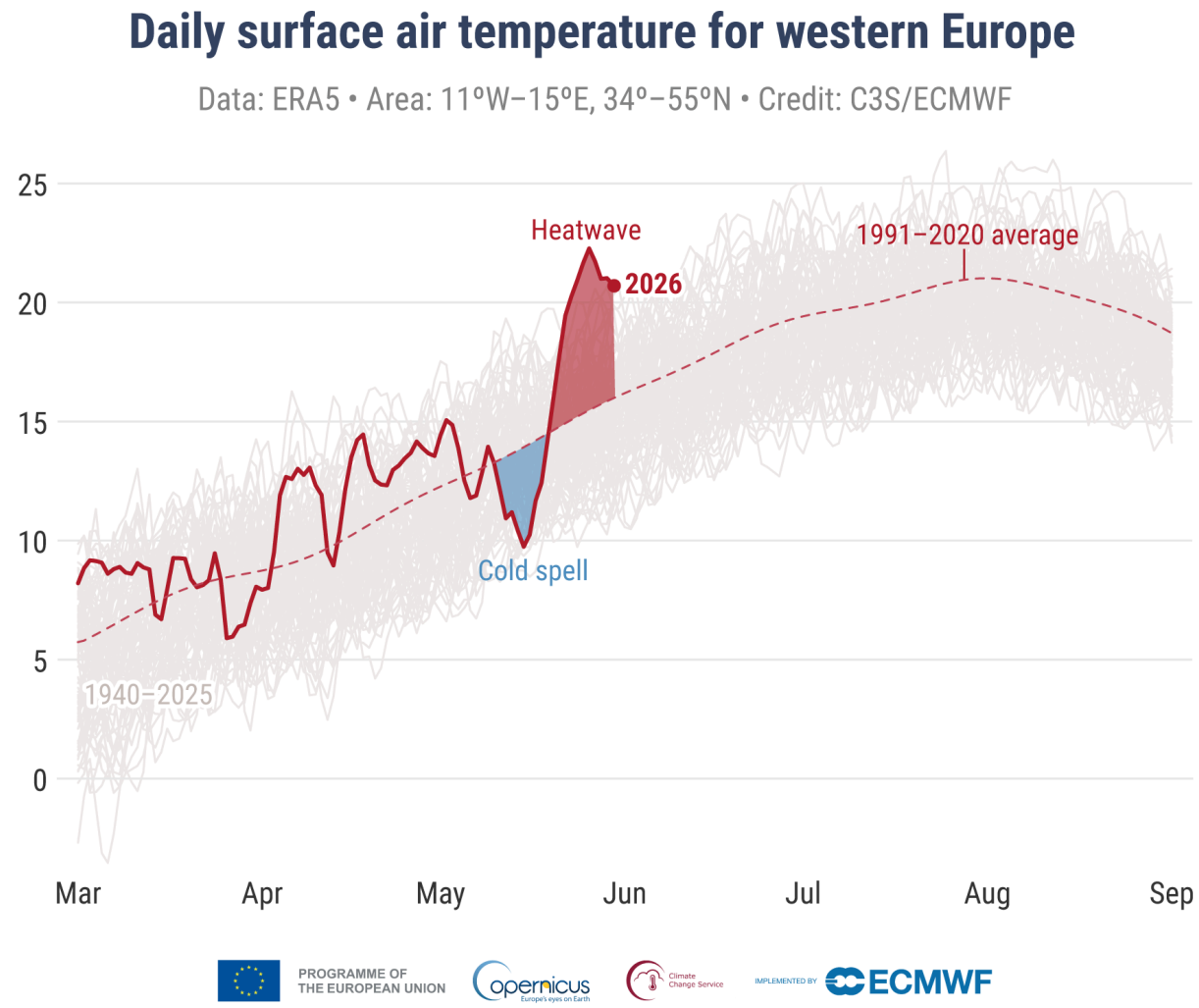
Task 2: News Visualisation

Find a recent visualisation about natural disasters (e.g., a specific storm, hurricanes, or related topics like wild fires). Post it to Canvas with a paragraph explaining:

- What it shows
- The data used
- Its original source
- What you like or dislike about it

Include constructive feedback and suggestions for improvement, and don't forget to add the URL to the original visualisation.

Figure 1: European Heatwave



Credit: https://climate.copernicus.eu/what-do-we-know-about-europes-early-and-intense-heatwave-may-2026

What it shows:

The graph shows the summer temperatures daily in Western Europe in degrees celsius, comparing it to the average of temperatures between 1991 and 2025 with the temperatures of years 1940 to 2025 in the background.

The Data's Source:

The data is from Copernicus' ERA5 Data Store, a European-centric data store tracking weather and temperature conditions.

My Opinion:

The graph very clearly visually lays out the summer average and the anomalous data from summer 2026 differently, allowing very easy visual comparison. It also labels the anomalies and what they are called in the "Cold Spell" and "Heatwave" notations. The

background of the graph also shows all previous summers graphs, giving a good visual indicator of the temperature variation and previous anomalous summers. The graph however does not label the measures of the axis and could further include an average of all temperatures since 1940.

Task 4: Create Visualisations in a Jupyter or Colab Notebook

Create 4 different data visualisations using datasets from Kaggle, BBC News, or any other useful source in a Jupyter Notebook or Google Colab to check your Python setup.

Use Python libraries such as Matplotlib, Seaborn, or others to create your visualisations. Each visualisation should use a different chart type or approach. Submit the notebook on Canvas along with a brief description of the data sources and the approach you took for each visualisation.

Visualisation 1: Buyers of New Homes

Source: Housing In London 2025 Dataset - London Data Store

Sheet 2.6 titles removed, converted to CSV.

https://data.london.gov.uk/dataset/housing-in-london-the-evidence-base-for-the-mayors-housing-strat-24rpx

```
#import libraries
import matplotlib.pyplot as plt
import pandas as pd

#setting data frame
df = pd.read_csv("london_new_home_buyers.csv")

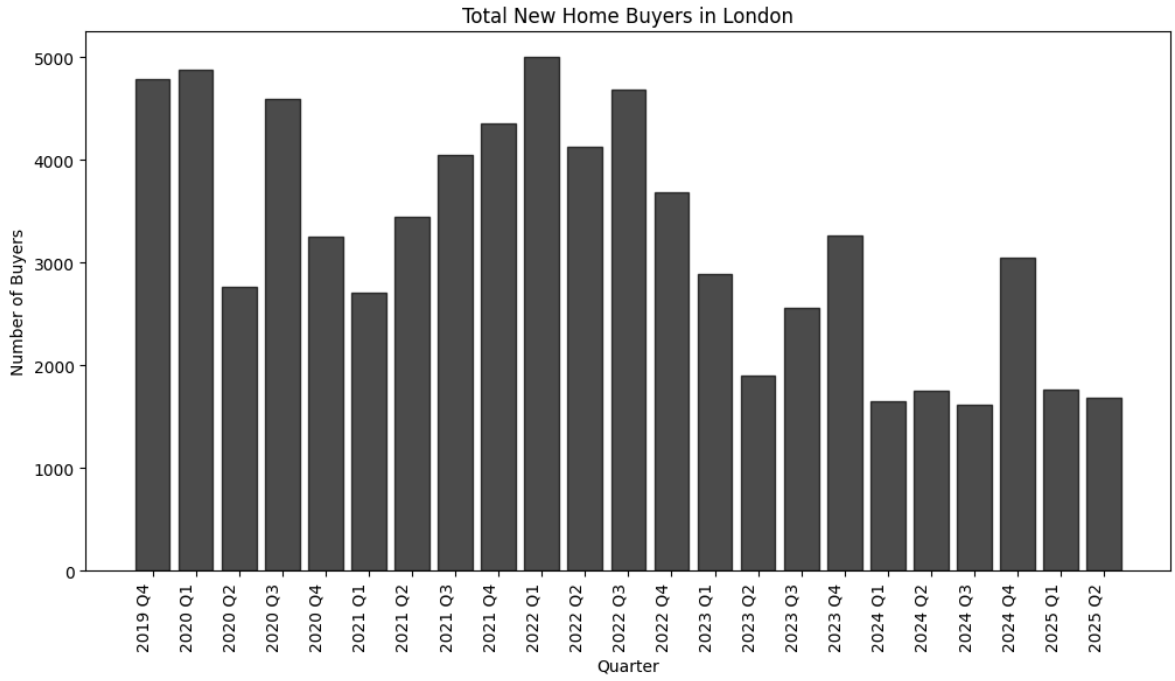

total_row = df[df['buyer_type'] == 'Total'].iloc[0]

#extract timeframes and values
quarters = df.columns[1:]
values = [total_row[q] for q in quarters]

#plot graph
plt.figure(figsize=(12, 6))
plt.bar(quarters, values, color='#000000', alpha=0.7, edgecolor='black')
plt.xlabel('Quarter')
plt.ylabel('Number of Buyers')
plt.title('Total New Home Buyers in London')
```

```
plt.xticks(rotation=90, ha='right')

plt.show()
```



Visualisation 2: Relative Poverty in London

Source: London Datastore

xls -> csv

https://data.london.gov.uk/dataset/hbai-poverty-in-london-e79lj

```
#import libraries
import matplotlib.pyplot as plt
import pandas as pd

#set dataframe
df = pd.read_csv("relative_poverty.csv", skiprows=4)
#Claude used to help learn how to reformat to be readable in line below
df.rename(columns={df.columns[0]: 'Year'}, inplace=True)


#defining the axis and focusing on stats for all people with before and aft
year = df['Year'].values
all_people_bhc = pd.to_numeric(df.iloc[:, 1], errors='coerce')
```

```
all_people_ahc = pd.to_numeric(df.iloc[:, 4], errors='coerce')

#plotting the graph
plt.figure(figsize=(14, 6))
plt.plot(year, all_people_bhc, color = 'g', marker='s', label='Before Housi
plt.plot(year, all_people_ahc, color = 'b', marker='s', label='After Housir
plt.xlabel('Period', fontsize=12)
plt.ylabel('Relative Poverty Rate (%)', fontsize=12)
plt.title('Relative Poverty: All People in UK', fontsize=14)
plt.xticks(rotation=90, ha='right')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```